

Abstract No.

Disrupting Golgi dependent signalling uncovers a Vulnerability in Medulloblastoma stem cells

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Background: Medulloblastoma (MB) is the most common malignant pediatric brain tumor, which frequently exhibits treatment resistance and recurrence, largely attributed to MB stem cells (MBSCs). Recently, we demonstrated that treatment of MBSCs with Brefeldin A (BFA), an inhibitor of Golgi function, reduces the stem-like phenotype by inducing the proteolytic cleavage of the stem cell marker CD133, thereby impairing its function. However, the limited clinical applicability of BFA has driven the search for structurally related compounds with greater target specificity. Among these, Golgicide A (GCA) has emerged as a potent inhibitor of Golgi function. Based on this rationale, we investigated the effects of GCA on MBSCs.

Methods: G3MB cell lines (D283 and D341) were grown in stem selective medium and treated with GCA for 48 hrs. Stemness and apoptotic proteins and oncogenic signalling levels were assessed by western blotting. Gene expression data were retrieved using R2 genomics platform.

Results: GCA impaired the stem-like phenotype of MBSCs by inducing proteolytic cleavage of CD133, reducing the expression of the stemness markers, and significantly decreasing medullosphere formation. Furthermore, GCA triggered apoptotic signaling and inhibited the PI3K/AKT/mTOR pathway. Notably, treatment with GCA also resulted in a marked downregulation of a Golgi-resident protein (Golgi-RP), previously implicated as an oncogenic factor in other tumoral contexts. Bioinformatic analyses further revealed that Golgi-RP is preferentially expressed in Group 3 MB, supporting its potential relevance in MB biology.

Conclusions: Our findings identify GCA as a structural and functional analogue of BFA capable of targeting CSC-associated pathways in MB. Modulation of CD133 processing and Golgi-dependent signaling emerges as a promising therapeutic strategy in MB, warranting further preclinical investigation. Building on these findings, an ongoing in silico and in vitro screening of structurally related natural and synthetic products is being conducted using GCA as hit compounds.



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